Lunbing (Caesar) Chen

Bio-inspired Flight · Dynamic Soaring · Aerodynamics · Autonomous Flight

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Research Profile

I am a Ph.D. candidate in J. C. Wu Center for Aerodynamics at Shanghai Jiao Tong University. My research spans **bio-inspired flight, dynamic soaring, aerodynamics, and autonomous flight**, with a focus on low-energy, long-endurance UAV flight in complex environmental flows. My doctoral research connects two complementary pathways: reducing energy expenditure through passive aerodynamic disturbance mitigation and harvesting atmospheric energy through closed-loop dynamic soaring. My first-author work has been published in the *Journal of Fluid Mechanics* [1], *Cell Reports Physical Science* [2], and *iScience* [3], with a manuscript under second-round revision at *Nature Communications* [4].

Research Interests

- **Dynamic soaring and autonomous environmental-energy flight**: energy–direction trade-offs, local sensing, state-feedback policies, and closed-loop UAV implementation [3, 4, 8].
- **Bio-inspired aerodynamics and disturbance-resilient flight**: passive disturbance mitigation, avian flight mechanisms, and environmental energy use [2, 9].
- **Unsteady aerodynamics and flow–structure interaction**: aerodynamic sensitivity, vortex dynamics, transient force generation, and reconfigurable structures [1, 5, 6, 10, 11].

Education

Shanghai Jiao Tong University Sep. 2022–Jun. 2027 (expected)
Ph.D. Candidate, J. C. Wu Center for Aerodynamics Shanghai, China

- Dissertation topic: *Passive Disturbance Mitigation and Closed-loop Dynamic Soaring for Bio-inspired Long-endurance Flight in Environmental Flows*
- Advisors: Prof. Yang Xiang and Prof. Hong Liu

Shanghai Jiao Tong University Sep. 2018–Jun. 2022
B.Eng., School of Aeronautics and Astronautics Shanghai, China

Research Experience

Mechanics and Feedback Control of Dynamic Soaring 2022–Present
Flight mechanics, optimal control, deep reinforcement learning

- Formulated 3-DOF glider dynamics and a direct-collocation optimal-control framework solved with IPOPT to map flight envelopes and quantify the allocation of harvested energy between sustained soaring and directional progress [3, 8].
- Built a continuous-control reinforcement-learning environment and trained maximum-entropy Soft Actor–Critic policies with curriculum learning and systematic observation/reward ablations to identify the local-sensing and state-feedback requirements of step-level dynamic soaring [4].

Closed-loop Dynamic Soaring UAV: Integration and Field Testing 2024–Present
UAV platform, state estimation, flight control, system integration

- Designed and integrated a modular sensing–decision–control architecture, combining onboard state and wind sensing, feedback control, DRL policy deployment, and real-time execution on a fixed-wing UAV.

- Built a Gazebo/PX4 simulation and embedded-integration workflow around a Raspberry Pi 5, and conducted field flight tests and manual fixed-wing piloting for platform validation and controller development.

Bio-inspired Aerodynamics and Passive Flight Robustness 2022–2026
Flow–structure interaction, passive reconfiguration, turbulence, aerodynamic sensitivity

- Designed towing-tank experiments on accelerating reconfigurable plates, combining programmable motion, precision force measurements, high-speed imaging, and time-resolved multi-plane PIV to resolve transient load redistribution and develop quasi-steady and flow-informed force models [1].
- Conducted grid-generated-turbulence wind-tunnel experiments on avian and planform-matched engineered wings using force measurements, hot-wire anemometry, multi-section SPIV, and flight-dynamic analysis to quantify aerodynamic sensitivity, passive stability, and control demand [2].
- Performed laser-sheet smoke visualization and high-speed imaging of live geese during stationary takeoff to connect wing kinematics with the evolving aerodynamic flow structures [9].

Unsteady Vortex Dynamics and Force Generation 2021–2025
Vortex rings, impulse theory, accelerating bodies, PIV, low-order force modeling

- Combined time-resolved PIV and synchronized force measurements to characterize vortex-ring formation, growth, and wake evolution in accelerating-body flows [1, 10].
- Performed three-dimensional transient simulations in ANSYS Fluent to analyze vortex-ring evolution in confined and asymmetric domains [11].
- Applied the vorticity-moment theorem and impulse-based force decomposition to connect measured vorticity evolution with transient force generation [1].

Earlier Aerospace Engineering Projects 2019–2022
Uncertainty quantification, numerical simulation, engineering software

- Developed a numerical framework for high-dimensional uncertainty quantification of thermochemical ablation in C/C nozzle throat inserts [7].
- Developed a mobile-based spacecraft management application for status monitoring and task publishing.

Publications

First-author / Corresponding-author Publications

*Corresponding author

[1] **Lunbing Chen**, Zhuoqi Li, Qilin Wu, Yufei Yin, Yang Xiang, and Hong Liu. "On the load-shifting phenomenon of accelerating reconfigurable circular plates". In: *Journal of Fluid Mechanics* 1029 (2026), A8. DOI: 10.1017/jfm.2026.11165.

[2] **Lunbing Chen**, Suyang Qin, Qilin Wu, Jinpeng Huang, Yufei Yin, Yong Chen, Yang Xiang, and Hong Liu. "Passive aerodynamic robustness reduces disturbance amplification in flight". In: *Cell Reports Physical Science* 7.10 (2026), p. 103581. DOI: 10.1016/j.xcrp.2026.103581.

[3] **Lunbing Chen**, Yufei Yin, Yang Xiang, Suyang Qin, and Hong Liu. "Optimal dynamic soaring trades off energy harvest and directional flight". In: *iScience* 28.6 (2025). DOI: 10.1016/j.isci.2025.112540.

[4] **Lunbing Chen**, Jixin Lu, Yufei Yin, Jinpeng Huang, Yong Chen, Yang Xiang, and Hong Liu. *Learning step-level dynamic soaring in shear flow*. Under second-round revision at *Nature Communications*. 2026. DOI: 10.48550/arXiv.2604.12413. arXiv: 2604.12413.

[5] Jinpeng Huang, Qilin Wu, **Lunbing Chen***, Yang Xiang, and Hong Liu. "Static Geometric Configuration and Vortex Ring Reshaping Dominate the Load-shifting of Accelerating Plates". Under review at *Physics of Fluids*. 2026.

[6] Qilin Wu**, Lunbing Chen***, Yang Xiang, and Hong Liu. "Shape Independence and Aerodynamic Truncation Effect of Low-Aspect-Ratio Accelerating Plates". In preparation for *Physical Review Fluids*. 2026.

[7] **Lunbing Chen**, Na Sun, Linying Li, and Bin Zhang. "High-Dimensional Uncertainty Quantification of Thermochemical Ablation for C/C Throat Insert". In: *Journal of Propulsion Technology* 43.3 (2022). (in Chinese), pp. 266–275. DOI: 10.13675/j.cnki.tjjs.200583.

Selected Co-authored Publications

[8] Yufei Yin**, Lunbing Chen**, Yang Xiang, Suyang Qin, and Hong Liu. "Flight Envelope and Energy Equilibrium Strategy of Sailplanes to Perform Dynamic Soaring". In: *International Conference on Aerospace System Science and Engineering*. Springer, 2024, pp. 17–27. DOI: 10.1007/978-981-96-2440-9_3.

[9] Jinpeng Huang, Yang Xiang**, Lunbing Chen**, Suyang Qin, Jixin Lu, Sen Ye, Xiaobin Huang, Yong Chen, and Hong Liu. "Geese achieve stationary takeoff via synergistic wing kinematics and enhanced aerodynamics". Submitted to *PNAS*. 2025. DOI: 10.48550/arXiv.2512.20894.

[10] Zhuoqi Li**, Lunbing Chen**, Yang Xiang, Hong Liu, and Fuxin Wang. "Two independent mechanisms with distinct laws for the generation of drag force on accelerating plates". In: *Physics of Fluids* 36.10 (2024). DOI: 10.1063/5.0227258.

[11] Yufei Yin**, Lunbing Chen**, Liwei Hu, Yang Xiang, Yumin Zhong, and Hong Liu. "Evolution and dynamics of vortex rings in a confined and asymmetric region". In: *Physics of Fluids* 37.12 (2025). **Editor's Pick, *Scilight***. DOI: 10.1063/5.0304637.

[12] Bin Yu**, Lunbing Chen**, Zhen Wang, Xunchen Liu, Bin Zhang, and Hong Liu. "Study on the structural characteristics of strong radial swirl flow based on Lagrangian and Eulerian vortex identification methods". In: *CCTAM 2021*. (in Chinese). 2021.

Technical Skills

UAV Systems and Flight Testing: PX4, Gazebo, Raspberry Pi 5, fixed-wing UAV integration, onboard sensing, manual piloting, field flight testing

Learning and Optimal Control: Reinforcement learning, PPO, SAC, Gym, curriculum learning, direct collocation, IPOPT, trajectory optimization

Experimental Aerodynamics: Towing-tank and wind-tunnel experiments, grid-generated turbulence, precision force measurement, PIV/SPIV, hot-wire anemometry, laser-sheet smoke visualization, high-speed imaging

CFD and Flight Dynamics: ANSYS Fluent, OpenFOAM, 3/6-DOF flight dynamics, ODE integration, stability analysis, vorticity-moment and impulse methods, low-order modeling

Scientific Programming: Python, MATLAB, Fortran, C++, Java, LaTeX, Git